Computer exercise 2 (mandatory)

STA310

> ! Written report
>
> This exercise should be submitted as a written report in Canvas. Use Quarto to produce a PDF. Relevant R-code, output, figures, tables, comments and interpretations should be included in the report. The schedueled session is for questions and discussion, but you will need to work on the exercise in your own time as well!

Relevant packages

```
suppressPackageStartupMessages(library(tidyverse))
library(survival)
```

(You are free to use additional packages as well.)

Kaplan-Meier survival

This is a simple function to estimate Kaplan-Meier survival.

```
#' Kaplan-Meier survival
#' @param time what?
#' @param status what?
#' @return what?
#' @examples
#' time <- c(2, 3, 5, 6, 8)
#' status <- c(1, 1, 0, 1, 0)
#' km(time, status)
#' survfit(Surv(time, status) ~ 1) |> summary()
km <- function(time, status) {
  ts <- sort(unique(time[status == 1]))
  S <- 1
  out <- data.frame(time = ts, n = NA, d = NA, S = NA)
  for (i in seq_along(ts)) {
    t <- ts[i]
    n <- sum(time >= t)
    d <- sum(time == t & status == 1)
    S <- S * (1 - d / n)
```

```
    out$n[i] <- n
    out$d[i] <- d
    out$S[i] <- S
  }
  out
}

# Vectorised version
km2 <- function(time, status) {
  t <- sort(unique(time[status == 1]))
  n <- sapply(t, function(z) sum(time >= z))
  d <- sapply(t, function(z) sum(time == z & status == 1))
  data.frame(time = t, n, d, S = cumprod(1 - d / n))
}
```

1. Run the example (see `@examples` above) to verify similarity between `km()`, `km2()` and the survival package equivalent. Are there any differences?

Include the function definition above (copy and paste either `km()` or `km2()` into a code block in your report) and:

2. Replace "what?" in the function description with relevant information

3. Add detailed comments within the function to explain what it does (step by step)

4. Modify the function to add 95 % confidence intervals to the point estimates. HINT: eq 2.11/2.12 in Collett, `?cumsum` and Wald CI.

5. Again, compare the results between `km()` (or `km2()`) and the survival package equivalent. Are they the same? If not, why? HINT: check argument `conf.type` in `?survfit.formula` (can you reproduce your results by changing this argument?).

Bonus: Re-define: `time <- c(2, 2, 1, 3, 2)` and `status <- c(1, 0, 1, 1, 0)` and rerun the analysis. What happened to the SE/CI? Why?

Kaplan-Meier plot

> **Tip**
>
> Reset `time` and `status` to the original example data if you have changed them in the previous bonus question.

Create a simple plot function:

```
plot_km <- function(x) ggplot(x, aes(time, S)) + geom_step()
```

Compare `plot_km(km(time, status))` and `plot(survfit(Surv(time, status) ~ 1), conf.int = FALSE)`.

6. How and why do the results differ (if they do)?

7. Explain (in words) how you would improve your functions (`km()`, `plot_km()` or both) to make your plot look like a real Kaplan-Meier plot.

Bonus: Implement your suggested changes from (7) and check the result.

Cox

Here is a simple Cox function for one covariate `x`:

```
cox <- function(time, status, x, beta = 0, tol = 1) {
  repeat {
    U <- I <- 0 # score and information
    for (i in which(status == 1)) {
      R <- which(time >= time[i]) # Risk set
      w <- exp(beta * x[R])
      mean_x <- sum(w * x[R]) / sum(w)
      mean_x2 <- sum(w * x[R]^2) / sum(w)
      U <- U + x[i] - mean_x # Score
      I <- I + mean_x2 - mean_x^2 # Information
    }
    beta_new <- beta + U / I # Newton-Raphson
    if (abs(beta_new - beta) < tol) {
      break
    }
    beta <- beta_new
  }

  c(beta = beta_new, SE = sqrt(1 / I))
}
```

> **Tip**
>
> Recapture section 3.3 from the book.

8. As for `km()` (`km2()`), include the function definition above (copy and paste into a code block in your report) and add a description of the function and comments within the function to explain what it does in more details.

Define a covariate `x <- c(0, 1, 0, 1, 1)`.

9. Compare `cox(time, status, x)` to the equivalent analysis performed by functions from the survival package.

10. Do the results differ? If so, why? HINT: what is the purpose of argument `tol` (you may find `?coxph.control` relevant)? Try different values of `tol` and check if the results from `cox()` and `coxph()` can reach agreement.

Analysis

We will use the `lung` data from the survival package. Read about it: `?lung`.

Try `with(lung, cox(time, status - 1, ph.karno))`.

11. Formulate a research question associated with the analysis (the code you tried).

12. Did you get any result? If not, why? HINT: perform the same analysis with the equivalent survival package function and read the output carefully

13. Fix the problem and check that the result from `cox()` is equivalent to the result from the equivalent survival package function.

14. Interpret the result and summarise your findings (using statistical, as well as clinical language).

15. How would you improve the analysis? You may reformulate the research question and you may use any available tool (methods/packages) for the analysis.

16. Perform your suggested analysis and present your results with relevant figures, tables and/or summaries/interpretations.